

Information Theory and Its Application in Deep Learning: Final Solutions

1. Differential Entropy and Maximum Entropy (30 points)

Let X be a continuous random variable. Recall $h(X) = -\int f(x) \log f(x) dx$ (nats).

- (a) Let $X \sim \text{Exp}(\lambda)$, i.e. $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Compute $h(X)$. (10 points)
- (b) Let $Y = cX$ for a constant $c > 0$. Find $h(Y)$ in two ways: (i) from the scaling rule, and (ii) by identifying the distribution of Y . (10 points)
- (c) Show that among *all* densities supported on $(0, \infty)$ with mean $\mathbb{E}[X] = \mu$, the exponential density $f(x) = \frac{1}{\mu} e^{-x/\mu}$ maximizes h . (10 points)

Solution: Differential Entropy and Maximum Entropy.

- (a) Take the logarithm of the density first: $\log f(x) = \log \lambda - \lambda x$. Substituting into the definition and splitting the integral,

$$\begin{aligned} h(X) &= -\int_0^\infty \lambda e^{-\lambda x} (\log \lambda - \lambda x) dx \\ &= -\log \lambda \underbrace{\int_0^\infty \lambda e^{-\lambda x} dx}_{=1} + \lambda \underbrace{\int_0^\infty x \lambda e^{-\lambda x} dx}_{=\mathbb{E}[X]}. \end{aligned}$$

The first integral is 1 because f is a density; the second is $\mathbb{E}[X] = \frac{1}{\lambda}$. Hence

$$h(X) = -\log \lambda + \lambda \cdot \frac{1}{\lambda} = 1 - \log \lambda.$$

- (b) **(i) Scaling rule.** For $Y = cX$ with $c > 0$ the density transforms as $f_Y(y) = \frac{1}{c} f_X(y/c)$, so

$$h(Y) = -\int f_Y(y) \log f_Y(y) dy = -\int \frac{1}{c} f_X(y/c) [\log f_X(y/c) - \log c] dy.$$

Substituting $x = y/c$ (so $dy = c dx$) gives $h(Y) = -\int f_X(x) [\log f_X(x) - \log c] dx = h(X) + \log c$. Therefore $h(Y) = 1 - \log \lambda + \log c$.

(ii) Identify the law. If $X \sim \text{Exp}(\lambda)$ then $Y = cX$ has $\Pr(Y \leq y) = \Pr(X \leq y/c) = 1 - e^{-\lambda y/c}$, i.e. $Y \sim \text{Exp}(\lambda/c)$. By part (a) with rate λ/c , $h(Y) = 1 - \log(\lambda/c) = 1 - \log \lambda + \log c$. The two answers agree.

- (c) Let g be any density on $(0, \infty)$ with $\int_0^\infty x g(x) dx = \mu$, and let $f(x) = \frac{1}{\mu} e^{-x/\mu}$ be the exponential with the same mean. Start from non-negativity of KL and expand the integrand:

$$0 \leq D(g||f) = \int g \log \frac{g}{f} = \int g \log g - \int g \log f = -h(g) - \int g(x) \log f(x) dx.$$

The key point is that $\log f$ is *linear* in x : $\log f(x) = -\log \mu - x/\mu$, so its g -average depends on g only through the mean,

$$\int g(x) \log f(x) dx = -\log \mu - \frac{1}{\mu} \int_0^\infty x g(x) dx = -\log \mu - 1,$$

which is the same value it takes under f itself. Substituting back, $0 \leq -h(g) + \log \mu + 1$, i.e.

$$h(g) \leq 1 + \log \mu.$$

The bound is attained by f , since $h(f) = 1 - \log(1/\mu) = 1 + \log \mu$ (part (a) with $\lambda = 1/\mu$). Hence the exponential maximizes h over this class, with maximum entropy $1 + \log \mu$, and equality holds iff $D(g||f) = 0$, i.e. $g = f$.

2. Rate–Distortion of an Independent Gaussian Pair (35 points)

Consider a vector source $\mathbf{X} = (X_1, X_2)$ with *independent* components $X_i \sim \mathcal{N}(0, \sigma_i^2)$, and squared-error distortion $d(\mathbf{x}, \hat{\mathbf{x}}) = \sum_{i=1}^2 (x_i - \hat{x}_i)^2$. Work with the information rate–distortion function

$$R(D) = \min_{p(\hat{\mathbf{x}}|\mathbf{x}): \mathbb{E}\|\mathbf{X} - \hat{\mathbf{X}}\|^2 \leq D} I(\mathbf{X}; \hat{\mathbf{X}}),$$

and write R_i for the single-component function $R_i(D_i) = \frac{1}{2} \log \frac{\sigma_i^2}{D_i}$ ($0 < D_i \leq \sigma_i^2$). Rates are in bits, $\log = \log_2$. The goal is to prove

$$R(D) = \min_{\substack{D_1, D_2 \geq 0 \\ D_1 + D_2 \leq D}} (R_1(D_1) + R_2(D_2)).$$

(a) **Decoupling lemma.** For *any* joint distribution of $(\mathbf{X}, \hat{\mathbf{X}})$, show

$$I(\mathbf{X}; \hat{\mathbf{X}}) \geq I(X_1; \hat{X}_1) + I(X_2; \hat{X}_2).$$

Hint: write $I = h(\mathbf{X}) - h(\mathbf{X} | \hat{\mathbf{X}})$; use independence on the first term, the chain rule on the second, and “conditioning reduces entropy.” (10 points)

(b) **Converse.** Show that

$$R(D) \geq \min_{D_1 + D_2 \leq D} (R_1(D_1) + R_2(D_2)).$$

Hint: take any test channel feasible for $R(D)$; set $D_i = \mathbb{E}[(X_i - \hat{X}_i)^2]$ so $D_1 + D_2 \leq D$. Apply part (a), then bound each $I(X_i; \hat{X}_i)$ below using the definition of the single-letter R_i . (10 points)

(c) **Achievability.** Show the matching upper bound

$$R(D) \leq \min_{D_1 + D_2 \leq D} (R_1(D_1) + R_2(D_2)),$$

so that equality holds. *Hint: fix any split (D_1, D_2) with $D_1 + D_2 \leq D$ and build a product test channel $p(\hat{\mathbf{x}} | \mathbf{x}) = \prod_i p_i^*(\hat{x}_i | x_i)$ from the per-component optimizers. Compute its mutual information and its distortion. (15 points)*

Solution: Rate–Distortion of an Independent Gaussian Pair.

(a) **(Decoupling lemma.)** Write the mutual information in differential-entropy form, $I(\mathbf{X}; \hat{\mathbf{X}}) = h(\mathbf{X}) - h(\mathbf{X} | \hat{\mathbf{X}})$, and treat the two terms separately.

First term. Since $X_1 \perp X_2$, the joint density factorizes and the entropy is additive: $h(\mathbf{X}) = h(X_1) + h(X_2)$.

Second term. By the chain rule for differential entropy,

$$h(\mathbf{X} | \hat{\mathbf{X}}) = h(X_1 | \hat{\mathbf{X}}) + h(X_2 | X_1, \hat{\mathbf{X}}).$$

Conditioning on more variables can only reduce differential entropy. Dropping variables from each conditioning set (keep only $\hat{X}_1$ in the first term, only $\hat{X}_2$ in the second) gives the upper bound

$$h(X_1 | \hat{\mathbf{X}}) \leq h(X_1 | \hat{X}_1), \quad h(X_2 | X_1, \hat{\mathbf{X}}) \leq h(X_2 | \hat{X}_2),$$

because $\hat{\mathbf{X}} = (\hat{X}_1, \hat{X}_2)$ contains $\hat{X}_1$, and $(X_1, \hat{\mathbf{X}})$ contains $\hat{X}_2$. Adding the two, $h(\mathbf{X} | \hat{\mathbf{X}}) \leq h(X_1 | \hat{X}_1) + h(X_2 | \hat{X}_2)$. Combining the two terms,

$$I(\mathbf{X}; \hat{\mathbf{X}}) = h(\mathbf{X}) - h(\mathbf{X} | \hat{\mathbf{X}}) \geq \sum_{i=1}^2 (h(X_i) - h(X_i | \hat{X}_i)) = \sum_{i=1}^2 I(X_i; \hat{X}_i). \quad \blacksquare$$

- (b) **(Converse.)** Let $p(\hat{\mathbf{x}} | \mathbf{x})$ be *any* test channel feasible for $R(D)$, and define $D_i := \mathbb{E}[(X_i - \hat{X}_i)^2]$ for the distortion it induces on each component. Summing, $D_1 + D_2 = \mathbb{E}\|\mathbf{X} - \hat{\mathbf{X}}\|^2 \leq D$, so (D_1, D_2) is a feasible split. The channel's marginal $X_i \rightarrow \hat{X}_i$ is itself a single-letter test channel achieving distortion D_i , and the scalar rate-distortion function $R_i(D_i)$ is by definition the *smallest* mutual information over all such channels; therefore $I(X_i; \hat{X}_i) \geq R_i(D_i)$. Chaining this with part (a),

$$I(\mathbf{X}; \hat{\mathbf{X}}) \stackrel{(a)}{\geq} \sum_i I(X_i; \hat{X}_i) \geq \sum_i R_i(D_i) \geq \min_{D_1 + D_2 \leq D} \sum_i R_i(D_i).$$

The last quantity is a constant independent of the chosen channel. Taking the minimum of the left-hand side over all feasible channels — which by definition is $R(D)$ — preserves the inequality:

$$R(D) \geq \min_{D_1 + D_2 \leq D} (R_1(D_1) + R_2(D_2)). \quad \blacksquare$$

- (c) **(Achievability.)** Fix any split (D_1, D_2) with $D_1 + D_2 \leq D$. For each component let $p_i^*(\hat{x}_i | x_i)$ be the optimal scalar Gaussian test channel attaining $I(X_i; \hat{X}_i) = R_i(D_i)$ at distortion D_i , and form the *product* channel

$$p(\hat{\mathbf{x}} | \mathbf{x}) = p_1^*(\hat{x}_1 | x_1) p_2^*(\hat{x}_2 | x_2).$$

Because both the source ($X_1 \perp X_2$) and the channel factorize, the two pairs $(X_1, \hat{X}_1)$ and $(X_2, \hat{X}_2)$ are mutually independent, so the joint law satisfies $p(\mathbf{x}, \hat{\mathbf{x}}) = \prod_i p(x_i, \hat{x}_i)$. Mutual information of independent pairs is additive,

$$\begin{aligned} I(\mathbf{X}; \hat{\mathbf{X}}) &= \mathbb{E} \log \frac{p(\mathbf{X}, \hat{\mathbf{X}})}{p(\mathbf{X}) p(\hat{\mathbf{X}})} = \sum_i \mathbb{E} \log \frac{p(X_i, \hat{X}_i)}{p(X_i) p(\hat{X}_i)} \\ &= \sum_i I(X_i; \hat{X}_i) = \sum_i R_i(D_i), \end{aligned}$$

and the distortion adds by linearity of expectation, $\mathbb{E}\|\mathbf{X} - \hat{\mathbf{X}}\|^2 = \sum_i \mathbb{E}[(X_i - \hat{X}_i)^2] = D_1 + D_2 \leq D$. This product channel is therefore feasible for $R(D)$, so $R(D) \leq \sum_i R_i(D_i)$. Since this holds for *every* feasible split, we may minimize the right-hand side:

$$R(D) \leq \min_{D_1+D_2 \leq D} \sum_i R_i(D_i).$$

Together with the converse in part (b), the two bounds coincide:

$$R(D) = \min_{D_1+D_2 \leq D} (R_1(D_1) + R_2(D_2)). \quad \blacksquare$$

Solving this convex program (Lagrange multipliers) yields the *reverse water-filling* allocation $D_i = \min(\theta, \sigma_i^2)$ — but that step is not required here.

3. The χ^2 -Divergence: From Estimator Variance to the Evidence (35 points)

The χ^2 -divergence is $\chi^2(P\|Q) = \int \frac{(p-q)^2}{q} = \int \frac{p^2}{q} - 1$. After establishing its basic properties, you will use it twice: once to lower-bound the variance of an estimator, and once to upper-bound an intractable log-evidence.

- (a) **Foundations.** Show $\chi^2(P\|Q) = \text{Var}_Q\left(\frac{dP}{dQ}\right)$, and that $D(P\|Q) \leq \chi^2(P\|Q)$. *Hint: $\log t \leq t - 1$. (10 points)*
- (b) **Hammersley–Chapman–Robbins.** For any statistic T with $\mathbb{E}_P[T] = a$, $\mathbb{E}_Q[T] = b$, prove

$$\text{Var}_Q(T) \geq \frac{(a - b)^2}{\chi^2(P\|Q)}.$$

Then, for a parametric family $\{P_\theta\}$ with an unbiased estimator of θ , take $Q = P_\theta$, $P = P_{\theta+h}$ and let $h \rightarrow 0$ to recover the Cramér–Rao bound $\text{Var}_\theta(T) \geq 1/I(\theta)$. *Hint: express $a - b$ as a covariance under Q and apply Cauchy–Schwarz. (15 points)*

- (c) **A χ^2 upper bound on the evidence (CUBO).** For a latent-variable model $p(x) = \int p(x, z) dz$, a proposal $q(z | x)$, and weight $w = \frac{p(x, z)}{q(z | x)}$, define $\text{CUBO}_2 = \frac{1}{2} \log \mathbb{E}_{q(z|x)}[w^2]$. Show $\log p(x) \leq \text{CUBO}_2$, that the gap is exactly $\frac{1}{2} \log(1 + \chi^2(p(z | x) \| q(z | x)))$, and hence $\text{ELBO} \leq \log p(x) \leq \text{CUBO}_2$. *Hint: $p(x) = \mathbb{E}_q[w]$ and $(\mathbb{E}_q[w])^2 \leq \mathbb{E}_q[w^2]$; then write $w = p(x) \frac{p(z|x)}{q(z|x)}$. (10 points)*

Solution: The χ^2 -Divergence.

- (a) **(Foundations.)** Let $W = \frac{dP}{dQ} = p/q$ be the likelihood ratio. Its first two Q -moments are

$$\mathbb{E}_Q[W] = \int q \frac{p}{q} = \int p = 1, \quad \mathbb{E}_Q[W^2] = \int q \left(\frac{p}{q}\right)^2 = \int \frac{p^2}{q}.$$

Hence $\text{Var}_Q(W) = \mathbb{E}_Q[W^2] - (\mathbb{E}_Q[W])^2 = \int \frac{p^2}{q} - 1 = \chi^2(P\|Q)$, the first claim. For the comparison with KL, apply $\log t \leq t - 1$ with $t = p/q$ pointwise:

$$D(P\|Q) = \int p \log \frac{p}{q} \leq \int p \left(\frac{p}{q} - 1\right) = \int \frac{p^2}{q} - \int p = \int \frac{p^2}{q} - 1 = \chi^2(P\|Q).$$

- (b) (**Hammersley–Chapman–Robbins.**) Write the mean gap as a Q -expectation against W . Since $\mathbb{E}_Q[W] = 1$,

$$\begin{aligned} a - b &= \mathbb{E}_P[T] - \mathbb{E}_Q[T] = \int T p - \int T q = \int T(p - q) \\ &= \int q T \left(\frac{p}{q} - 1 \right) = \mathbb{E}_Q[T(W - 1)]. \end{aligned}$$

Because $\mathbb{E}_Q[W - 1] = 0$, this expectation is exactly a covariance: $a - b = \mathbb{E}_Q[T(W - 1)] = \text{Cov}_Q(T, W)$. By the Cauchy–Schwarz inequality for covariance,

$$(a - b)^2 = \text{Cov}_Q(T, W)^2 \leq \text{Var}_Q(T) \text{Var}_Q(W) = \text{Var}_Q(T) \chi^2(P \| Q),$$

and rearranging gives $\text{Var}_Q(T) \geq (a - b)^2 / \chi^2(P \| Q)$.

Cramér–Rao limit. Put $Q = P_\theta$, $P = P_{\theta+h}$. Unbiasedness means $\mathbb{E}_{\theta'}[T] = \theta'$, so $a = \theta + h$, $b = \theta$, and $a - b = h$; thus $\text{Var}_\theta(T) \geq h^2 / \chi^2(P_{\theta+h} \| P_\theta)$. Taylor-expanding the density, $p_{\theta+h} = p_\theta + h \partial_\theta p_\theta + O(h^2)$, so $(p_{\theta+h} - p_\theta)^2 = h^2 (\partial_\theta p_\theta)^2 + O(h^3)$ and

$$\chi^2(P_{\theta+h} \| P_\theta) = \int \frac{(p_{\theta+h} - p_\theta)^2}{p_\theta} = h^2 \int \frac{(\partial_\theta p_\theta)^2}{p_\theta} + O(h^3) = h^2 I(\theta) + O(h^3),$$

where $I(\theta) = \int (\partial_\theta \log p_\theta)^2 p_\theta$ is the Fisher information. Letting $h \rightarrow 0$,

$$\text{Var}_\theta(T) \geq \frac{h^2}{h^2 I(\theta) + O(h^3)} \longrightarrow \frac{1}{I(\theta)}.$$

So HCR is a finite-difference Cramér–Rao bound, with χ^2 in the role of (squared) Fisher distance. ■

- (c) (**CUBO.**) First, $p(x)$ is the q -average of the weight:

$$p(x) = \int p(x, z) dz = \int q(z | x) \frac{p(x, z)}{q(z | x)} dz = \mathbb{E}_q[w].$$

Since the square is convex, Jensen gives $p(x)^2 = (\mathbb{E}_q[w])^2 \leq \mathbb{E}_q[w^2]$; taking $\frac{1}{2} \log$ (increasing) of both sides yields $\log p(x) \leq \frac{1}{2} \log \mathbb{E}_q[w^2] = \text{CUBO}_2$.

Exact gap. Factor the weight through the true posterior, $w = \frac{p(x, z)}{q(z | x)} = p(x) \frac{p(z | x)}{q(z | x)}$ (using $p(x, z) = p(x) p(z | x)$). Then

$$\begin{aligned} \mathbb{E}_q[w^2] &= p(x)^2 \mathbb{E}_q \left[\left(\frac{p(z | x)}{q(z | x)} \right)^2 \right] = p(x)^2 \int q(z | x) \frac{p(z | x)^2}{q(z | x)^2} dz \\ &= p(x)^2 \int \frac{p(z | x)^2}{q(z | x)} dz. \end{aligned}$$

By the foundation identity $\int \frac{p^2}{q} = 1 + \chi^2(p \| q)$ from part (a), the last integral is $1 + \chi^2(p(z | x) \| q(z | x))$, so

$$\text{CUBO}_2 = \frac{1}{2} \log \mathbb{E}_q[w^2] = \log p(x) + \frac{1}{2} \log(1 + \chi^2(p(z | x) \| q(z | x))).$$

The gap $\frac{1}{2} \log(1 + \chi^2) \geq 0$, and vanishes iff $\chi^2 = 0$, i.e. $q(z | x) = p(z | x)$. Finally recall the ELBO is $\mathbb{E}_q[\log w] \leq \log \mathbb{E}_q[w] = \log p(x)$ (Jensen, concavity of $\log$). Combining the two,

$$\text{ELBO} \leq \log p(x) \leq \text{CUBO}_2,$$

bracketing the intractable log-evidence between two computable bounds. The ELBO gap is the *reverse* KL $D(q(z | x) \| p(z | x))$ (mode-seeking), while the CUBO gap is driven by $\chi^2(p(z | x) \| q(z | x))$ (mass-covering). ■